

IZMAIL, Ukraine

WMO: 338890

Lat: 45.3703N

Lon: 28.8508E

Elev: 30

StdP: 100.97

Time Zone: 2.00 (EUE)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-10.0	-16.2	0.9	-11.7	-13.6	1.2	-9.5	13.5	-2.6	12.0	-2.4	3.0	0	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	32.8	21.3	31.4	20.8	29.8	20.3	22.6	29.6	21.9	28.8	21.2	27.9	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.5	15.3	25.5	19.7	14.4	24.8	18.8	13.7	24.3	66.6	29.5	63.9	29.0	61.4	28.0	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.8	-15.6	35.5	2.4	2.2	-17.3	37.1	-18.7	38.5	-20.1	39.7	-21.8	41.3		
(5)			-16.1	24.2	2.4	1.1	-17.8	24.9	-19.2	25.6	-20.5	26.2	-22.3	26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	-0.7	1.8	5.8	11.4	16.9	20.9	23.6	22.8	17.6	11.8	7.1	0.1		
	DBStd	9.35	4.73	5.18	4.20	3.73	3.22	2.63	2.50	3.06	3.51	4.29	4.71	5.02		
	HDD10.0	1184	333	231	143	27	1	0	0	0	1	29	111	308		
	HDD18.3	2893	591	463	390	208	65	7	1	2	55	207	338	566		
	CDD10.0	1778	0	1	11	71	216	326	421	398	227	86	22	1		
	CDD18.3	447	0	0	0	2	21	83	163	142	32	4	0	0		
	CDH23.3	4080	0	0	1	21	203	692	1573	1351	214	25	0	0		
	CDH26.7	1372	0	0	0	3	30	174	603	524	36	1	0	0		
(14) Wind	WSAvg	3.3	3.1	3.4	4.0	3.7	3.2	3.4	3.2	3.0	3.1	3.0	3.0	3.2		
(15) Precipitation	PrecAvg	478	31	37	32	35	46	57	50	39	46	27	38	41		
	PrecMax	778	163	120	95	87	127	146	179	108	155	90	114	161		
	PrecMin	298	0	3	0	5	5	4	0	4	2	2	2	2		
	PrecStd	100	29	28	27	19	31	32	37	29	45	20	28	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.4	21.2	26.1	29.4	32.3	34.8	35.1	29.7	26.1	20.3	12.8		
		MCWB	8.0	10.7	12.5	17.0	18.8	20.5	22.0	22.2	19.9	18.3	14.3	9.8		
	2%	DB	8.9	13.2	17.3	22.7	27.2	30.3	32.9	33.1	27.5	23.1	17.3	10.4		
		MCWB	6.2	8.9	10.6	14.8	18.0	20.3	21.5	21.1	18.6	17.1	12.9	8.0		
	5%	DB	7.0	10.8	14.7	20.0	25.6	28.8	31.6	31.4	25.9	20.4	15.1	8.3		
		MCWB	5.3	7.2	9.2	13.3	17.2	19.8	21.2	20.7	18.2	15.6	11.8	6.3		
	10%	DB	5.4	8.4	12.3	18.1	23.9	27.2	30.1	29.7	24.0	18.3	13.5	6.5		
		MCWB	3.9	5.9	7.9	12.1	16.5	19.0	20.6	20.0	17.3	14.4	11.0	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	11.2	12.8	17.6	20.5	22.3	23.8	24.3	21.4	19.6	15.0	10.7		
		MCDB	11.6	15.6	19.9	25.2	27.4	28.6	31.5	32.3	26.3	24.0	18.8	12.0		
	2%	WB	6.5	9.0	11.0	15.3	18.8	21.4	22.7	22.7	20.2	17.8	13.6	8.1		
		MCDB	8.3	12.9	16.5	21.5	25.0	27.9	30.3	29.0	25.3	21.8	16.6	10.0		
	5%	WB	5.3	7.5	9.6	14.0	17.9	20.6	22.0	21.8	19.1	16.3	12.4	6.4		
		MCDB	6.8	10.3	14.0	18.6	23.8	27.2	29.4	28.5	23.5	19.9	14.4	8.1		
	10%	WB	4.0	6.3	8.3	12.9	17.1	19.7	21.3	21.0	18.1	14.8	11.4	5.0		
		MCDB	5.4	8.3	11.6	17.2	22.8	25.8	28.5	27.4	22.7	17.8	13.3	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	7.3	8.0	9.5	10.9	10.6	11.1	10.9	9.9	8.8	6.6	6.0		
		MCDBR	7.9	10.8	12.1	13.0	13.5	12.9	13.4	13.5	12.3	11.6	10.0	8.3		
	5% WB	MCWBR	5.8	6.9	6.7	6.7	6.2	5.6	5.1	5.3	5.5	6.1	6.0	6.1		
		MCWBR	7.4	10.4	11.3	11.5	12.3	11.4	11.9	11.6	10.9	10.8	8.4	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.371	0.403	0.441	0.418	0.417	0.437	0.446	0.412	0.386	0.363	0.341		
		taud	2.298	2.236	2.196	2.118	2.266	2.296	2.253	2.218	2.279	2.328	2.334	2.322		
	Ebn at Noon		744	791	817	818	853	853	829	805	800	766	713	701		
		Edh at Noon	82	105	125	147	131	128	132	132	114	95	79	73		
(44) All-Sky Solar Radiation	RadAvg		1.21	2.00	3.29	4.78	6.05	6.42	6.41	5.75	4.17	2.62	1.38	1.03		
	RadStd		0.20	0.28	0.30	0.43	0.39	0.37	0.35	0.30	0.34	0.23	0.20	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	+70

Nomenclature: See separate page